

Modern Physics

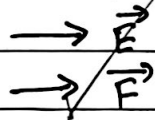
Electron

- > Charge of $e^- = -1.6 \times 10^{-19} \text{ C}$
- > mass of $e^- = 9.1 \times 10^{-31} \text{ kg}$
- > mass of proton = $1.6725 \times 10^{-27} \text{ kg}$
- > Specific charge of $e^- \left(\frac{e}{m} \right) = 1.76 \times 10^{11} \text{ C/kg}$

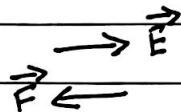
- Force acting on charge present in electric field

→ If charge ' q ' is placed in electric field ' E ', then force experienced by it is
$$\vec{F} = q\vec{E}$$

If $q = +ve$, $\vec{E}$ and $\vec{F}$ are in same direction



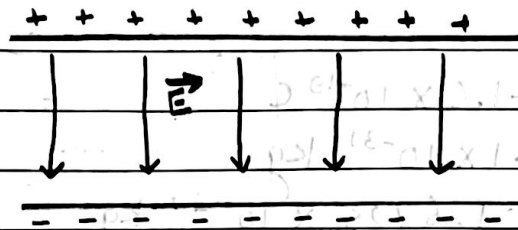
If $q = -ve$, $\vec{E}$ and $\vec{F}$ are in opposite direction.



Q. An electron is projected along the direction of electric field with certain velocity ' v '. Then

- It gets accelerated
- it gets retarded
- it moves with constant velocity

The acceleration of charge particle moving in electric field (a) = $\frac{F}{m} = \frac{qE}{m}$



Let V = potential difference between plates
 d = distance between plates

Then, electric field (E) = $\frac{V}{d}$

Q. An electron is present in electric field of 3000 V per m. find time taken by electron to travel 1 cm from rest.

Ans. Soln,

$$a = \frac{qE}{m}$$

$$= \frac{1.6 \times 10^{-19} \times 3000}{9.1 \times 10^{-31}}$$

$$= 5.27 \times 10^{14} \text{ m/s}^2$$

Again

$$s = ut + \frac{1}{2}at^2$$

$$s = \frac{1}{2}at^2$$

$$\text{or, } t = \sqrt{\frac{0.01 \times 2}{5.27 \times 10^{14}}}$$

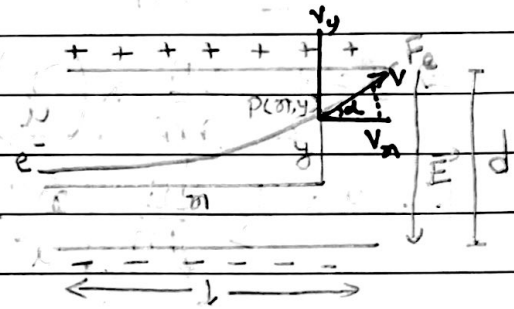
$$\therefore t = 6.16 \times 10^{-9} \text{ s}$$

✶ Motion of e^- in transverse electric field:

Let V = potential difference between plates

d = distance between plates

L = length of each plate



Let's consider an electron beam is projected with velocity ' u ' in direction perpendicular to $\vec{E}$. As soon as e^- enters the field, electric force $F_e = eE$ acts in vertical direction.

Let $P(x, y)$ be the position of electron at any time ' t '. Since no force acts on e^- in horizontal direction, motion is uniform ($a=0$). The horizontal distance ' x ' covered by e^- in time ' t ' is

$$x = u_x t + \frac{1}{2} a_x t^2 \rightarrow 0$$

$$x = u t$$

$$\Rightarrow \frac{x}{u} = t \rightarrow (i)$$

Since there is no initial motion along vertical direction, $u_y = 0$, the vertical distance ' y ' covered by e^- in time ' t ' is

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$y = \frac{1}{2} \cdot \frac{F}{m} \cdot t^2 \quad \left[\because a = \frac{F}{m} \right]$$

$$\Rightarrow y = \frac{1}{2} \frac{eE}{m} \times \left(\frac{x}{u} \right)^2 \rightarrow \text{From eqn (i)}$$

$$\therefore y = \frac{1}{2} \frac{eE}{m} \frac{x^2}{u^2}$$

$$y \propto x^2$$

The above equation shows that the nature of path followed by e^- in transverse $\vec{E}$ is parabolic.

Again,

Let 'v' be the resultant velocity of e^- at position P and V_x, V_y be its horizontal and Vertical components.

Now,

$$V_x = u + at \rightarrow 0$$

$$V_x = u$$

Again,

$$V_y = 0 + at$$

$$V_y = \frac{F}{m} \times t$$

$$V_y = \frac{eE}{m} \times \frac{x}{u}$$

The resultant velocity 'v' is given by

$$V = \sqrt{V_x^2 + V_y^2}$$

$$= \sqrt{u^2 + \left(\frac{eE}{m} \cdot \frac{n}{u} \right)^2}$$

Let ' α ' be the angle made by resultant velocity with horizontal direction,

Then,

$$\tan \alpha = \frac{V_y}{V_x} = \frac{\frac{eE}{m} \cdot \frac{n}{u}}{u} = \frac{eEn}{mu^2}$$

$$\therefore \alpha = \tan^{-1} \left(\frac{eEn}{mu^2} \right)$$

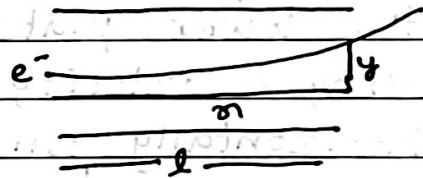
Numerical concepts:

- e^- emerge out of plates

In this case,

$$n = l$$

At the point of emergence,



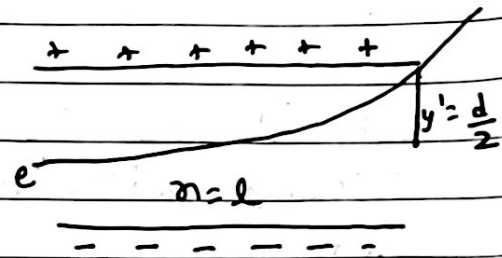
$$\hookrightarrow \text{Vertical deflection (y')} = \frac{1}{2} \frac{eE}{m} \cdot \frac{l^2}{u^2}$$

$$\hookrightarrow \text{Velocity (V)} = \sqrt{u^2 + \left(\frac{eE}{m} \cdot \frac{l}{u} \right)^2}$$

$$\hookrightarrow \alpha = \tan^{-1} \left(\frac{eEl}{mu^2} \right)$$

- e^- emerge by grazing the edge of plate

In this case, $x=l$, $y'=\frac{d}{2}$



$$\hookrightarrow y' = \frac{1}{2} \frac{eE}{m} \cdot \frac{l^2}{u^2} = \frac{d}{2}$$

$$\hookrightarrow v = \sqrt{u^2 + \left(\frac{eEl}{m} \cdot \frac{1}{u} \right)^2}$$

$$\hookrightarrow \alpha = \tan^{-1} \left(\frac{eEl}{mu^2} \right)$$

Q. Two plane metal plates 4 cm long are held horizontally 3 cm apart in a vacuum, one being vertically above the other. The upper plate is at a potential of 300 V and the lower plate is earthed. Electrons having a velocity of 10^7 m/s are injected horizontally midway between the plates and in a direction parallel to the 4 cm edge. Calculate the vertical deflection of the electron beam as it emerges from the plates. (e/m of $e^- = 1.8 \times 10^{18}$ C/kg).

→ Soln,

$$l = 4 \text{ cm} = 0.04 \text{ m}$$

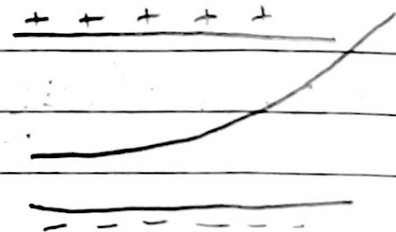
$$d = 3 \text{ cm} = 0.03 \text{ m}$$

$$V = 300 \text{ V}$$

$$u = 10^7 \text{ m/s}$$

$$y' = ?$$

Now,



$$E = \frac{V}{d} = \frac{300}{0.03}$$

$$= 10000 \text{ V/m}$$

$$y' = \frac{1}{2} \frac{eE}{m} \cdot \frac{l^2}{u^2}$$

$$= \frac{1}{2} \times 1.8 \times 10^{-18} \times 10000 \times \frac{(0.04)^2}{(10^7)^2}$$

$$= 0.0144 \text{ m}$$

Q. A beam of electrons, moving with a velocity of 10^7 m/s , enters midway between two horizontal parallel plates in a direction parallel to the plates. Each plate is 5 cm long. These plates are kept 2 cm apart and P.d of 90 V is applied between them. Calculate the velocity of the electron beam with it just grazes the edge of the positive plate.

→ Soln,

$$u = 10^7 \text{ m/s}$$

$$l = 5 \text{ cm} = 0.05 \text{ m}$$

$$d = 2 \text{ cm} = 0.02 \text{ m}$$

$$V = 90 \text{ V}$$

We have,

$$E = \frac{V}{d} = \frac{90}{0.02}$$

$$= 4500 \text{ V/m}$$

Now,

$$v = \sqrt{u^2 + \left(\frac{eE}{m} \cdot \frac{l}{u} \right)^2}$$

$$= \sqrt{(10^7)^2 + \left(\frac{1.8 \times 10^{11} \times 4500 \times 0.05}{10^7} \right)^2}$$

$$= \sqrt{1 \times 10^{14} + 1.64025 \times 10^{13}}$$

$$= \sqrt{1.164025 \times 10^{14}}$$

$$= 10788999.03 \text{ m/s}$$

$$= 2.99 \text{ km/hr.}$$

Q. A beam of electrons, moving with a velocity of 10^6 m/s enters midway between two horizontal parallel plates in a direction parallel to the plates. Each plate is 4 cm long. These plates are kept 2 cm apart and a potential difference V is applied between them. Calculate V if the beam is deflected so that it just grazes the edge of positive plate. (Given $\frac{e}{m}$ of electron = $1.8 \times 10^{11} \text{ C/kg}$).

→ Soln,

$$u = 10^6 \text{ m/s}$$

$$l = 4 \text{ cm} = 0.04 \text{ m}$$

$$d = 2 \text{ cm} = 0.02 \text{ m}$$

$$\frac{e}{m} = 1.8 \times 10^{11} \text{ C/kg.}$$

Now,

$$E = \frac{V}{d}$$

$$E = \frac{V}{0.02} \rightarrow (i)$$

$$V = 0.02 E \rightarrow (ii)$$

Again,

$$y' = \frac{d}{2} = \frac{1}{2} \cdot \frac{eE}{m} \left(\frac{d}{u} \right)^2$$

$$\frac{0.02}{2} = \frac{1}{2} \times \cancel{0.02} \cdot 1.8 \times 10^{11} \times \frac{V}{0.02} \times \left(\frac{0.04}{10^6} \right)^2$$

$$\Rightarrow 0.02 = 9 \times 10^{12} V \times 1.6 \times 10^{-15}$$

$$\Rightarrow 0.02 = 0.0144 V$$

$$\Rightarrow V = \frac{0.02}{0.0144}$$

$$V = 1.38 \text{ V}$$

$$\therefore V = 1.38 \text{ V}$$

Q. Electrons with kinetic energy of 500 eV are projected between two parallel charged metal plates in a direction parallel to the surface. The plates are at a p.d of 5V and 5mm apart and 4cm long. What is the angle the emergent beam makes with the original direction?

→ Soln,

$$K.E = 500 \text{ eV} = 500 \times 1.6 \times 10^{-19} \text{ J}$$

$$\frac{1}{2} m u^2 = 8 \times 10^{-17} \text{ J}$$

$$\Rightarrow \frac{1}{2} \times 9.1 \times 10^{-31} \times u^2 = 8 \times 10^{-17}$$

$$\Rightarrow u^2 = \frac{8 \times 10^{-17}}{4.55 \times 10^{-31}} = 1.75 \times 10^{14}$$

$$V = 5 \text{ V}$$

$$d = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$$

$$E = \frac{V}{d} = \frac{5}{5 \times 10^{-3}} = 10^3 \text{ V/m}$$

Now,

$$\alpha = \tan^{-1} \left(\frac{eE}{m} \times \frac{l}{u^2} \right)$$

$$\alpha = \tan^{-1} \left(\frac{1.8 \times 10^{11} \times 10^3 \times 0.04}{1.76 \times 10^{24}} \right)$$

$$\alpha = \tan^{-1} (0.0409)$$

$$\alpha \approx 2.34^\circ$$

$$\therefore \alpha = 2.34^\circ$$

Q Beams of electrons and protons having the same initial K.E enter normally into an electric field, which beam will be more curved?

→ If both have same initial K.E

We have,

$$y = \frac{1}{2} \cdot \frac{qE}{m} \cdot \frac{x^2}{u^2}$$

Again,

$$K.E = \frac{1}{2} mu^2$$

$$2K.E = mu^2$$

Now,

$$y = \frac{1}{2} \frac{qEx^2}{2K} = \frac{qEx^2}{4K} \quad [\because K.E = K]$$

Here,

for e^- and proton, q, E, x and K are all constants, so $y = \text{constant}$
i.e. both are equally curved.

II. If both e^- and p^+ have same initial velocity

We have,

$$y = \frac{1}{2} \frac{qE}{m} \frac{x^2}{u^2}$$

for e^- and p^+ , q, E, x and u all are constants

$$y \propto \frac{1}{m}$$

$\therefore m_p > m_e$, $y_p < y_e$ i.e. Path of electron will be more curved.

III. If both have same initial momentum

We have,

$$\begin{aligned} y &= \frac{1}{2} \frac{qE}{m} \cdot \frac{x^2}{u^2} \\ &= \frac{qEx^2}{2m^2u^2} \times m \\ &= \frac{qEx^2 \cdot m}{2p^2} \end{aligned}$$

for e^- and p^+ , q, E, x^2, p are constants,

$$y \propto m$$

$\therefore m_p > m_e$, $y_p > y_e$.

i.e. Path of proton will be more curved.

• Force acting on charge moving in $\vec{B}$

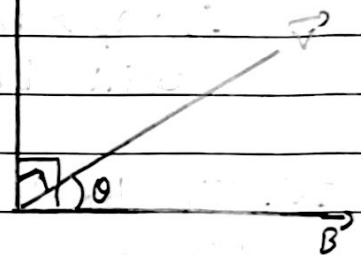
→ If charge 'q' is moving with velocity 'v' making angle ' θ ' with magnetic field, the force acting on it is given by

$$F = Bqv \sin \theta \quad [\theta = \text{angle between } \vec{v} \text{ and } \vec{B}] \Rightarrow$$

In vector form,

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Direction of $\vec{F}$ is $\perp$ to the plane containing $\vec{v}$ and $\vec{B}$ ($\vec{F} \perp \vec{v}$, $\vec{F} \perp \vec{B}$)



→ Work done by Magnetic force (W) = zero

From work-Energy theorem,

$$W = \Delta K.E = 0$$

$$K.E = \text{constant}, \text{ speed} = \text{constant}$$

Q: A charge particle is moving inside magnetic field $\vec{B}$. Which of the following quantities changes?

a) speed

b) Energy

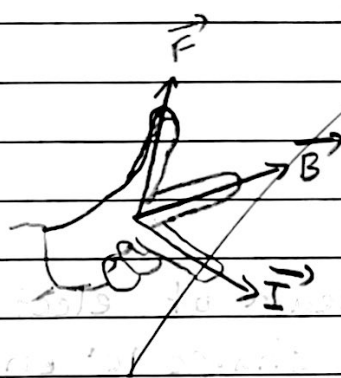
c) direction of motion ✓

d) ~~None~~ All of the above

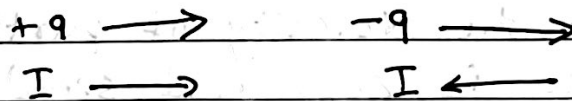
⇒ In electric field Ans (d)

- Direction of $\vec{F}$ is given by Fleming's Left hand Rule

→ According to this rule, "If the fore finger, middle finger and thumb of left hand are extended so as to make them mutually perpendicular, such that the fore finger points the direction of field and middle finger points the direction of conventional current, then the thumb indicates the direction of force.



Note: Direction of current is along the direction of positive charge and opposite to the direction of negative charge.



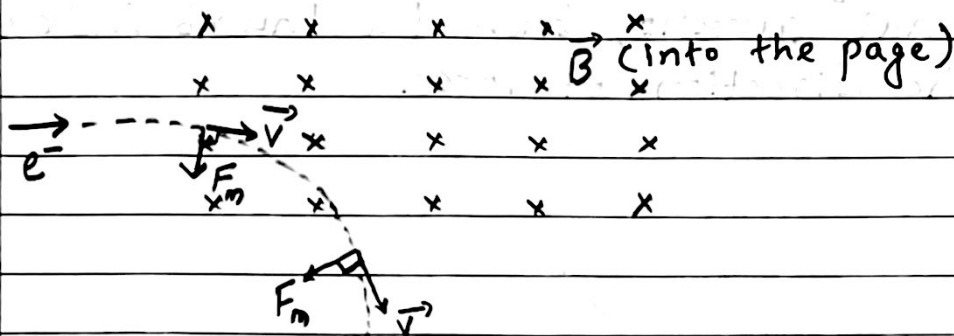
Q An electron is moving toward east and magnetic field is directed toward north then force acting on it is directed

- a) West c) Vertically Inward
b) South d) Vertically downward.

Q. A proton is moving along vertically upward direction and magnetic field is directed toward ~~so~~ west. Then force acting on it is directed toward

Ans South

- Motion of electron in transverse magnetic field.



Let us consider a beam of electron each having mass ' m ' and charge ' e ' enters with velocity ' v ' at a right angle to the direction of magnetic field.

Since magnetic force acts continuously perpendicular to the velocity of particle, work done by force is zero and speed of particle remains constant. So, its direction changes and the path of particle becomes circular. Here, for circular motion necessary centripetal force is provided by magnetic force.

i.e. Magnetic force = Centripetal force.

$$\Rightarrow e v B \sin \theta = \frac{m v^2}{r}$$

Here, $\theta = 90^\circ$,

$$e v B = \frac{m v^2}{r}$$

$$r = \frac{m v}{e B}$$

$$r = \frac{P}{e B} \quad [\because P = \text{momentum}]$$

$$r = \frac{\sqrt{2 m K}}{e B} \quad K = \text{kinetic energy}$$

$$\begin{aligned} K &= \frac{1}{2} m v^2 \\ &= \frac{1}{2} \frac{m^2 v^2}{m} \\ K &= \frac{P^2}{2m} \\ P^2 &= 2mK \\ P &= \sqrt{2mK} \end{aligned}$$

Q. Find the ratio of radius of circular paths when electron and proton are projected at an right angle to the direction of magnetic field and have same momentum.

→ Soln

WE have,

$$r = \frac{m v}{e B} = \frac{P}{e B}$$

For e^- and p^+ , p , e and B are all same,

$$\frac{V_e}{V_p} = \frac{1}{1}$$

Q. An electron having energy 500 eV enters at right angle to a uniform magnetic field of 10^{-4} T. If its specific charge is 1.75×10^{11} C/kg, calculate the radius of its orbit.

→ Soln,

$$K.E = 500 \text{ eV}$$

$$\Rightarrow \frac{1}{2} mv^2 = 500 \text{ eV}$$

$$\Rightarrow v^2 = 1000 \times \frac{e}{m}$$

$$\Rightarrow v^2 = 1000 \times 1.75 \times 10^{11} \text{ C/kg}$$

$$\Rightarrow v^2 = 1.75 \times 10^{14} \text{ C/kg}$$

$$\Rightarrow v = \sqrt{1.75 \times 10^{14} \text{ C/kg}}$$

$$\Rightarrow v = 13228756.56$$

Now,

$$r = \frac{mv}{eB} = \frac{v}{\frac{e}{m} \times B}$$

$$= \frac{13228756.56}{1.75 \times 10^{11} \times 10^{-4}}$$

$$= 0.75 \text{ m}$$

Note: If charge 'q' is accelerated through p.d of 'V' volt from rest

$$qV = \frac{1}{2}mv^2$$

Date _____
Page _____

Q. A beam of proton is accelerated from rest through a p.d of 2000 V and then enters a uniform magnetic field which is perpendicular to the direction of the proton beam. If the flux density is 0.4 T, calculate the radius of the path which the beam describes. (mass of proton = 1.7×10^{-27} kg).

→ Soln.

Since proton beam is accelerated through p.d of 2000 V.

$$qV = \frac{1}{2}mv^2$$

$$\Rightarrow 1.6 \times 10^{-19} \times 2000 = \frac{1}{2} \times 1.7 \times 10^{-27} \times v^2$$

$$\Rightarrow v^2 = \frac{4000 \times 1.6 \times 10^{-19}}{1.7 \times 10^{-27}}$$

$$\Rightarrow v^2 = 3.76 \times 10^{11}$$

$$\Rightarrow v = \sqrt{3.76 \times 10^{11}}$$

$$\Rightarrow v = 613571.99 \text{ m/s}$$

Now,

$$r = \frac{mv}{qB}$$

$$qB$$

$$= \frac{1.7 \times 10^{-27} \times 613571.99}{1.6 \times 10^{-19} \times 0.4}$$

$$= 0.016 \text{ m}$$

$$= 1.6 \text{ cm}$$

• We have,

$$r = \frac{mv}{eB}$$

Time period of revolution is given by

$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \cdot \frac{mv}{eB} = 2\pi \left(\frac{m}{eB} \right)$$

$$= \frac{2\pi m}{eB}$$

Now,

$$\text{frequency of revolution } (f) = \frac{1}{T} = \frac{eB}{2\pi m}$$

$$\text{Angular frequency } (\omega) = 2\pi f = 2\pi \cdot \frac{eB}{2\pi m} = \frac{eB}{m}$$

Note: f , ω , and T depends on magnitude of charge, mass and strength of magnetic field.

They do not depend on speed.

Q. An electron moves in a circular path of radius 20 cm in a uniform magnetic field of 2 mT. find the speed of electron and period of revolution.

→ Soln,

$$B = 2 \text{ mT} = 2 \times 10^{-3} \text{ T}$$

$$r = 20 \text{ cm} = 0.2 \text{ m}$$

$$r = \frac{mv}{eB}$$

$$0.2 = \frac{9.1 \times 10^{-31} \times v}{1.6 \times 10^{-19} \times 2 \times 10^{-3}}$$

$$\Rightarrow v = \frac{0.2 \times 1.6 \times 10^{-19} \times 2 \times 10^{-3}}{9.1 \times 10^{-31}}$$

$$v = 70329670.3 \text{ m/s}$$

$$\therefore v = 7.03 \times 10^7 \text{ m/s}$$

Again,

$$T = \frac{2\pi m}{eB}$$

$$= \frac{2\pi \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19} \times 2 \times 10^{-3}}$$

$$= 5.6875 \times 10^{-9} \text{ s}$$

$$= 1.78 \times 10^{-8} \text{ s}$$

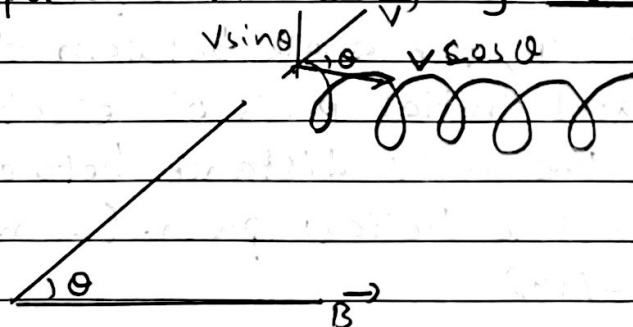
• Concept : $F = Bqv \sin \theta$

→ If $\theta = 0^\circ, 180^\circ, F = 0$

- charge particle moves along straight path without being deviated.

→ If $\theta \neq 0^\circ, 180^\circ, 90^\circ$,

- charge particle moves along helical path.



→ Helical Path is due to superposition of straight motion and circular motion.

> radius of helical path (r) = $\frac{m(v \sin \theta)}{eB}$

> Time period (T) = $\frac{2\pi}{\omega} = \frac{2\pi}{v/r} = 2\pi \left(\frac{m v \sin \theta}{eB} \right) \frac{1}{v \sin \theta}$

$$= \frac{2\pi m}{eB}$$

> pitch of helical path (P):

→ The linear distance travelled by charge particle in one revolution is called pitch. It is given by

$$P = \text{velocity} \times t$$

$$= v \cos \theta \times T$$

$$= v \cos \theta \times \frac{2\pi m}{eB}$$

$$\therefore P = \frac{2\pi m v \cos \theta}{eB}$$

Q. An electron with a velocity of 10^7 m/s enters a region of uniform magnetic flux density of 0.1 T, the angle between the direction of field and initial path of electron being 25° . Find the axial distance between two turns of the helical path. (e/m of electron = 1.8×10^{11} C/kg)

→ Soln,

$$V = 10^7 \text{ m/s}$$

$$B = 0.1 \text{ T}$$

$$\frac{e}{m} = 1.8 \times 10^{11} \text{ C/kg}$$

$$\theta = 25^\circ$$

$$V_x = V \sin \theta$$

$$= 10^7 \sin 25^\circ$$

$$V_y = V \cos \theta$$

$$= 10^7 \cos 25^\circ$$

$$\approx 9.063 \times 10^6 \text{ m/s}$$

$$T = \frac{2\pi m}{eB} = \frac{2\pi}{\left(\frac{e}{m}\right)B}$$

$$= \frac{2\pi}{1.8 \times 10^{11} \times 0.1}$$

$$\approx 3.491 \times 10^{-10}$$

Now,

$$P = V_y \times T$$

$$= 9.063 \times 10^6 \times 3.491 \times 10^{-10}$$

$$\approx 0.003164 \text{ m}$$

$$= 3.16 \times 10^{-3} \text{ m}$$

Q. An electron and proton are projected at an right angle to the direction of magnetic field and have same kinetic energy. Whose trajectory is more curved.

→ In magnetic field, $r \propto \frac{1}{\text{curvature}}$

(i) If both have same K.E

We have,

$$r = \frac{\sqrt{2mk}}{qB}$$

for e^- and p^+ , q , B and k are same

$$r \propto \sqrt{m}$$

$$\therefore m_p > m_e, r_p > r_e$$

Then $C_p < C_e$

i.e. path of e^- will be more curved.

(ii) If both have same momentum,

$$r = \frac{p}{qB}$$

for e^- and p^+ , p , q and B are same

then,

$$r_p = r_e$$

$$C_p = C_e$$

i.e. path of both are equally curved.

(iii) If both have same velocity.

We have

$$r = \frac{mv}{qB}$$

for p^+ and e^- , v , q , and B are all constant.

$$r \propto m$$

$$\therefore m_p > m_e, r_p > r_e$$

$$\text{Then, } C_p < C_e$$

i.e. path of e^- will be more curved.

• Cross Field

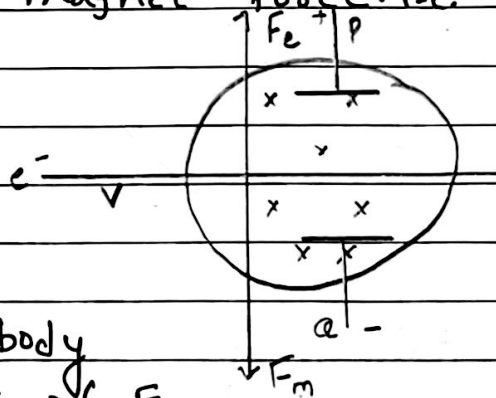
→ A region of uniform electric field and magnetic field applied simultaneously perpendicular to each other such that a charge particle entering normally into this region passes undeviated is called cross field.

Since deflection produced on charge particle due to electric force is equal to deflection produced due to magnet force. i.e.

$$F_e = F_m$$

$$eE = evB$$

$$v = \frac{E}{B}$$



i.e. Velocity of body is equal to ratio of E and B .

✓ J.J. Thompson's experiment: Determination of specific charge of electron in 1897.

> principle

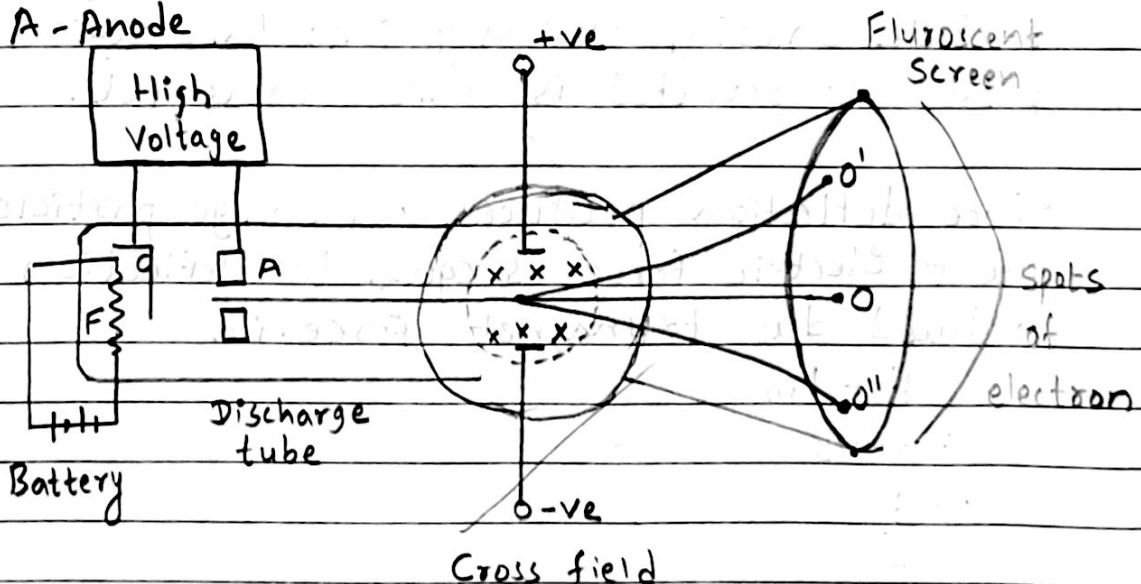
- Thompson used the principle of cross field to determine specific charge of electron.

The experimental set up of J.J. Thomson experiment is shown in fig:

F - filament

C - cathode

A - Anode



> Experimental setup:

- A highly evacuated discharge tube contains a cathode (C), anode (A), and a fluorescent screen.
- A high voltage (V) accelerates electrons from the cathode to the anode. Parallel plates create a uniform electric field (E).

- A magnetic field (B) is applied perpendicular to the electric field and electron beam. The fluorescent screen shows the position of the electron beam.

> Observation

- With no field: spot appears at O .
- With electric field only: Beam deflects to O' .
- With magnetic field only: Beam deflects to O'' .
- With both E and B field adjusted: Beam becomes undeflected and returns to O .

When uniform electric field is applied. The force acting on e^- due to electric field is

$$F_e = eE \rightarrow \textcircled{i}$$

When a uniform magnetic field ' B ' is applied perpendicular to beam, The force acting on e^- due to magnetic field is

$$F_m = evB \rightarrow \textcircled{ii}$$

When both fields are applied simultaneously, then beam is undeflected and strike screen at O .

$$\text{so, } F_e = F_m$$

$$\Rightarrow eE = eVB$$

$$\Rightarrow E = VB$$

$$\Rightarrow v = \frac{E}{B} \rightarrow (iii)$$

If P.d between cathode and anode is 'V', then

$$eV = \frac{1}{2} mv^2$$

$$\Rightarrow \frac{e}{m} = \frac{v^2}{2V} = \frac{E^2}{2VB^2}$$

If 'V' is the P.d between plates P and Q and 'd' be the distance between plates, then

$$\frac{e}{m} = \frac{V'^2}{2VB^2d^2} \left[\because E = \frac{V'}{d} \right]$$

Knowing the values of V, V', B, d, value of $\frac{e}{m}$ can be ~~determine~~ determined

which is found to be equal to $1.76 \times 10^{11} \text{ C/kg}$.

Q. A beam of electrons is under potential difference of $1.36 \times 10^4 \text{ V}$ applied across two parallel plates 4 cm apart and a magnetic field 2 mT at right angle to each other. If two fields produce no deflection in the electronic beam, calculate

- i) the velocity of electrons
- ii) the radius of the orbit in which the

beam will move, if the electric field is made zero.

→ Soln,

$$V = 1.36 \times 10^4 \text{ V}$$

$$d = 4 \text{ cm} = 0.04 \text{ m}$$

$$B = 2 \text{ mT} = 2 \times 10^{-3} \text{ T}$$

$$i) E = \frac{V}{d} = \frac{1.36 \times 10^4}{0.04} = 340000$$

$$V = \frac{E}{B} = \frac{340000}{2 \times 10^{-3}} = 1.7 \times 10^8 \text{ m/s}^2$$

$$ii) r = \frac{mv}{eB} = \frac{9.1 \times 10^{-31} \times 1.7 \times 10^8}{1.6 \times 10^{-19} \times 2 \times 10^{-3}} = 0.484 \text{ m}$$

- Millikan's oil drop experiment; Determination of charge of electron

Principle: It is based on Stoke's law in viscosity. i.e. When spherical drop of radius 'r' is falling through viscous medium of coefficient of viscosity 'η' and under action of many forces attain uniform velocity called terminal velocity, then net force acting on it is

$$F_v = 6\pi r\eta v$$

The experimental set up for MODE is shown in fig:

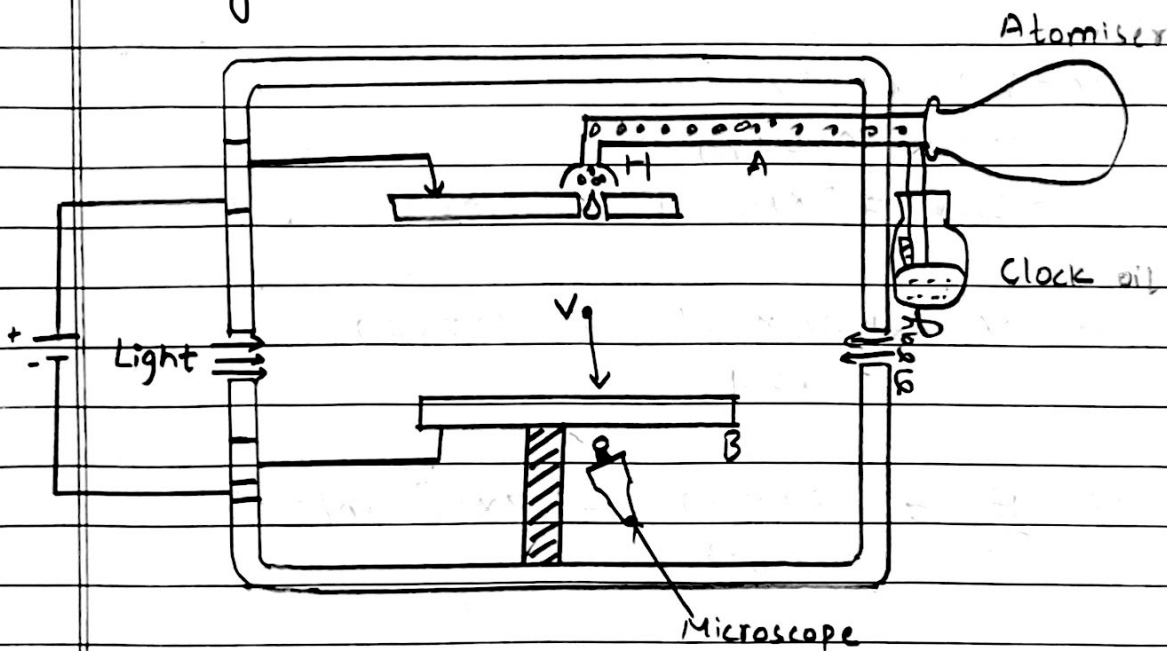


fig-Millikan's oil drop experiment

- Two parallel metal plates are connected to a high voltage source.
- Fine oil droplets of clock oil are sprayed between the plates through a small hole in the upper plate.
- X-rays ionize the air, causing electrons to attach to the oil droplets and give them charge.
- Light illuminates the droplets, and their motion is observed through a microscope. By adjusting the electric field, the charge on a droplet can be measured.

Let r = radius of drop under observation

σ = density of medium

ρ = density of oil drop

η = coefficient of viscosity of medium

➤ When electric field is switched off:

- When electric field is switched off, oil drop is found to move downward with terminal velocity (v_t) under following forces.
- Weight of drop acting downward ($W = mg$)
- Upthrust acting upward (U)
- ~~the viscous~~ Viscous force acting upward (F_v).

In equilibrium,

$$F_v + U = W$$

$$F_v = W - U$$

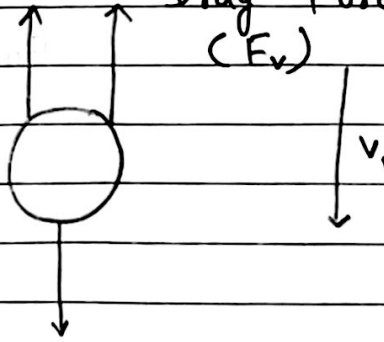
$$6\pi\eta r v_t = \frac{4}{3}\pi r^3(\rho - \sigma)g$$

~~an solving~~ $r = \sqrt{\frac{9\eta v_t}{2(\rho - \sigma)g}}$

➤ Without electric Field (falling)

Buoyancy
Force (U)

Drag Force
(F_v)



Weight (W)

→ When electric field is switched on

→ When electric field is switched on, oil drop is found to move upward with terminal velocity V_2 under the action of following forces:

→ Weight of drop acting downward ($W = mg$)

→ Upthrust acting in upward direction (U)

→ Electric Force ($F_e = qE$) acting in upward direction.

→ Viscous force acting in downward direction (F'_v)

In equilibrium,

$$F_e + U = F'_v + W$$

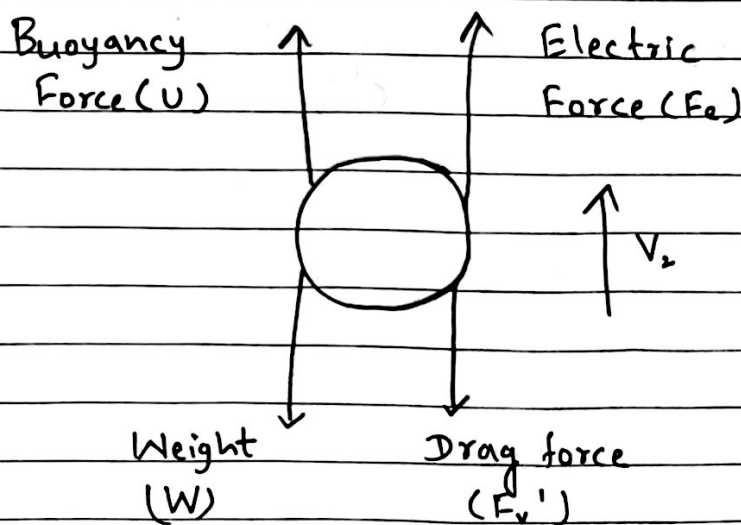
$$qE = 6\pi\eta r v_2 + (W - U)$$

$$qE = 6\pi\eta r v_2 + 6\pi\eta r v_1$$

$$qE = 6\pi\eta r (v_1 + v_2)$$

$$q = \frac{6\pi\eta r}{E} \sqrt{\frac{9\eta v_1}{2(\rho - \sigma)g}} (v_1 + v_2)$$

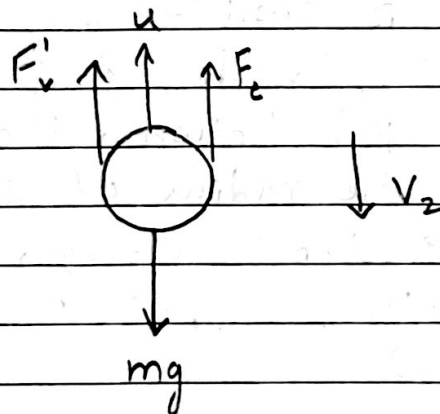
- With electric Field (rising)



knowing the values of ρ , σ and η , V_1 , V_2 .
Values of q on the drop can be calculated.
Repeating the experiment for many drop-
lets, Millikan found that all charges turned
out to be an integral multiple of unique
value of $1.6 \times 10^{-19} \text{ C}$ which is the smallest
charge existing in nature. On this basis,
Millikan concluded that the basic charge
i.e. charge of electron is $1.6 \times 10^{-19} \text{ C}$.

Note: If charge drop is allowed to move
downward with terminal velocity V_2 in
presence of electric field then above
equation becomes.

$$q = \frac{6\pi\eta}{E} \sqrt{\frac{9\eta V_1}{2(\rho - \sigma)g}} \quad (V_1 = V_2)$$

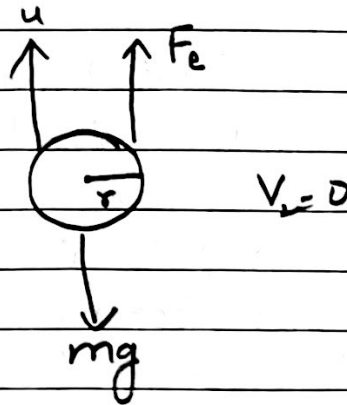


In his later experiment, he used the bal-
anced drop method i.e. if drop remains
stationary in presence of electric field then
 $V_2 = 0$.

Then above equation becomes,

$$q = \frac{6\pi\eta}{E} \sqrt{\frac{9\eta v_c}{2(\rho - \sigma)g}} v_c$$

Stationary:



Q. In a Millikan-type apparatus the horizontal plates are 1.5 cm apart. With the electric field switched off, an oil drop is observed to fall with the steady velocity 2.5×10^{-2} cm/s. When the electric field is switched on, the upper plate being positive, the drop just remains stationary when the p.d between the plates is 1500 V. Calculate the radius of the drop and the number of electronic charges.

(Given: $\rho_{oil} = 900 \text{ kg/m}^3$, $\eta_{air} = 1.8 \times 10^{-5} \text{ Nsm}^{-2}$, Neglect air density)

4 Soln,

Here, $d = 1.5 \text{ cm} = 1.5 \times 10^{-2} \text{ m}$

$V_c = 2.5 \times 10^{-4} \text{ m/s}$

$V = 1500 \text{ V}$

Since air density is neglected, no upthrust act

We have,

$$r = \sqrt{\frac{9\eta v}{2(\rho - \sigma)g}}$$

$$r = \sqrt{\frac{9 \times 1.8 \times 10^{-5} \times 2.5 \times 10^{-4}}{2 \times 900 \times 9.8}}$$

$$r = 1.5 \times 10^{-6}$$

Now,

$$F_e = mg$$

$$\Rightarrow qE = mg$$

$$\Rightarrow ne \cdot E = \frac{4}{3} \pi r^3 \times \rho \times g \rightarrow (1)$$

$$\Rightarrow E = \frac{V}{d} = \frac{1500}{1.5 \times 10^{-2}} = 1 \times 10^5$$

Then,

$$n \times 1.6 \times 10^{-19} \times 1 \times 10^5 = \frac{4}{3} \times \frac{22}{7} \times (1.5 \times 10^{-6})^3 \times 900 \times 9.8$$

$$n = \frac{1.2474 \times 10^{-13}}{1.6 \times 10^{-19} \times 1 \times 10^5}$$

$$= 7.79$$

$$\therefore n \approx 8$$

Q. In the millikan's oil drop experiment, a drop is observed to fall with a terminal velocity 1.4 mm/s in the absence of electric field. When a vertical electric field of 4.9×10^5 V/m is applied, the droplet is observed to continue to move downward at a lower terminal speed 1.21 mm/s. Calculate the charge on the

drop. ($\rho_{\text{oil}} = 750 \text{ kg/m}^3$, $\sigma_{\text{air}} = 1.29 \text{ kg/m}^3$
 $\eta_{\text{air}} = 1.81 \times 10^{-5} \text{ kg/ms}$).

→ Soln,

$$V_1 = 1.4 \text{ mm/s} = 1.4 \times 10^{-3} \text{ m/s}$$

$$E = 4.9 \times 10^5 \text{ V/m}$$

$$V_2 = 1.21 \text{ mm/s} = 1.21 \times 10^{-3} \text{ m/s}$$

$$q = ?$$

We know that,

$$\begin{aligned} q &= \frac{6\pi\eta}{E} \sqrt{\frac{9\eta V_1}{2(\rho - \sigma)g}} (V_1 - V_2) \\ &= \frac{6 \times 3.14 \times 1.81 \times 10^{-5}}{4.9 \times 10^5} \sqrt{\frac{9 \times 1.81 \times 10^{-5} \times 1.4 \times 10^{-3}}{2(750 - 1.29) \times 9.8}} \\ &\quad \times (1.4 \times 10^{-3} - 1.21 \times 10^{-3}) \\ &= 6.95 \times 10^{-10} \times \sqrt{1.55 \times 10^{-11}} \times 0.19 \times 10^{-3} \\ &= 5.19 \times 10^{-19} \\ \therefore q &= 5.19 \times 10^{-19} . \end{aligned}$$

Q. Calculate the radius of water drop which would be just remain suspended in an electric field of 300 V/m and charge with one electron.

→ Soln,

$$qE = mg$$

$$\Rightarrow \frac{1.6 \times 10^{-19}}{1.6} \times 300 = m$$

$$\Rightarrow m = 4.8 \times 10^{-18}$$

Now,

$$m = V \times \rho$$

$$m = \frac{4}{3} \pi r^3 \times 1000$$

$$4.8 \times 10^{-18} = \frac{4}{3} \times \frac{22}{7} \times r^3 \times 1000$$

$$r^3 = 1.14 \times 10^{-21}$$

$$r = 1.04 \times 10^{-7} \text{ m}$$

* Short Questions:

① Which type of liquid is used in Millikan's oil drop experiment and why?

Ans. Non-Volatile and viscous type liquid
i.e. clock oil is used in MDE because such type of liquid has low vapour pressure which reduces the problem of evaporation and it gives quite small droplets.

Note: Water is not used because water has high vapour pressure and can easily evaporate.

② Millikan's experiment cannot be done with bigger drops. Why?

→ For bigger oil drops, high electric field is required to achieve terminal velocity and to apply such field large plates are required which is practically impossible. So small drops are used.

③ What is the importance of Millikan's oil drop experiment?

→ It gives information of quantization of charge

→ It is used to determine mass of electron.

4) A charged particle moves through a region of space with constant velocity. If the external magnetic field in this region is zero, can we conclude that the external electric field in the region is also zero? explain.

→ Yes, electric field in this region must be zero because if electric field exist in region, electric force ($F = qE$) acts on it and its velocity cannot remain constant.

* Note: Electric Field ਜਾਂਦਾ ਜਾਂਦਾ ਹੈ, ਜਾਂਦਾ condition on ਜਾਂਦਾ velocity change ਹੁੰਦਾ ।

• Discussion Questions

Short Questions.

(i) Compare the specific charge of an electron with that of proton.

$$\begin{aligned} \rightarrow \frac{S_e}{S_p} &= \frac{e/m_e}{e/m_p} \\ &= \frac{m_p}{m_e} = \frac{1.67 \times 10^{-27}}{9.1 \times 10^{-31}} \\ \therefore S_e &> S_p \end{aligned}$$

(5)



$$d \leq l$$

$$r \leq l$$

$$\frac{2mv}{qB} \leq l$$

Yes, it is possible if diameter of circular path formed is less than or equal to size of cubical size.

③ Why is a magnetic field used to deflect electron beam but not an electric field in a T.V. picture tube?

→ A magnetic field can produce a much larger deflecting force on fast-moving electrons than an electric field of practical magnitude, without requiring very high voltages. Magnetic deflection coils are also easier to construct, more compact, and give better focusing and linearity of deflection, so they are preferred over electric deflection plates in T.V. picture tubes.

⑤ A charged particle is fired into a cubical region of space where there is uniform magnetic field. Outside this region, there is no magnetic field, is it possible that the particle will remain inside the cubical region? Explain.

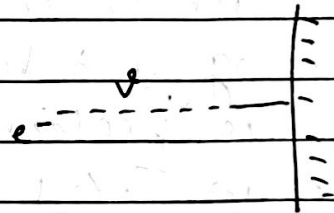
→ Yes, it is possible, provided the diameter of the circular path ($2r = \frac{2mv}{qB}$) is less than or

equal to the side length of the cubical region, and the particle enters in such a way that the entire circular path lies within the cube.

- 9) An electron beam after being accelerated from rest through a potential difference of 5 kV in vacuum is allowed to impinge normally on a fixed surface. If the incident current is 50 μ A, determine the force exerted on the surface assuming that it bring the electrons to rest. Take mass of e^- as 9.1×10^{-31} kg.

→ Since e^- beam is accelerated through p.d of 5 kV,

$$eV = \frac{1}{2} mv^2$$



$$\Rightarrow 1.6 \times 10^{-19} \times 5000 = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$$

$$\Rightarrow \frac{8 \times 10^{-16} \times 2}{9.1 \times 10^{-31}} = v^2$$

$$\Rightarrow v = \sqrt{1.758241758} \times 10^{13} \text{ m/s}$$

Now,

Let 'n' be the no. of e^- in beam

$$I = \frac{q}{t}$$

$$\Rightarrow I = \frac{ne}{t}$$

$$\Rightarrow 50 \times 10^{-6} = \frac{n \times 1.6 \times 10^{-19}}{t}$$

$$\Rightarrow \frac{n}{t} = 3.125 \times 10^{14}$$

Then,

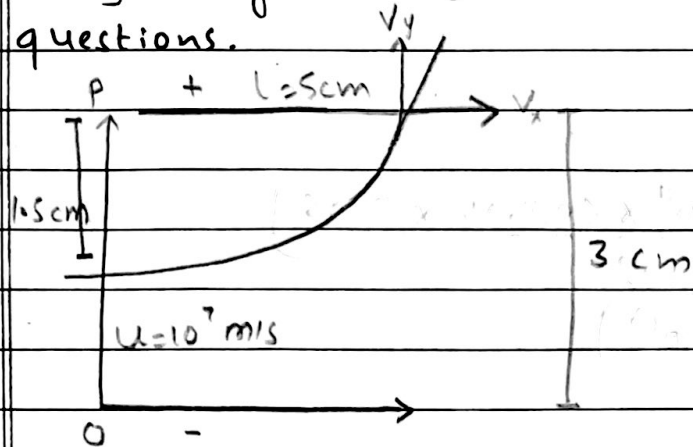
$$F = \frac{\text{change in momentum}}{\text{time}}$$

$$= \frac{n m_e (v - 0)}{t}$$

$$= 3.125 \times 10^{14} \times 9.1 \times 10^{-31} \times 41931393.47$$

$$= 1.192 \times 10^{-8} \text{ A}$$

3. Study the given figure and answer the given questions.



Calculate

a) potential difference between plates.

→ Soln,

$$u = 10^7 \text{ m/s}$$

$$d = 3 \text{ cm} = 0.03 \text{ m}$$

$$y = \frac{d}{2} = 1.5 \text{ cm} = 0.015 \text{ m}$$

Now,

$$y = \frac{1}{2} \times \frac{eE}{m} \left(\frac{L}{u} \right)^2$$

$$y = \frac{1}{2} \times 1.76 \times 10^{11} \times E \times \left(\frac{L}{u} \right)^2$$

$$0.015 = \frac{1}{2} \times 1.76 \times 10^{11} \times E \times \left(\frac{0.05}{10^7} \right)^2$$

$$E = \frac{0.015 \times 2 \times 10^{14}}{1.76 \times 10^{11} \times (0.05)^2}$$

$$E = 6818.18$$

Now,

$$\frac{V}{d} = \frac{E}{r}$$

$$\text{or, } V = 6818.18 \times 0.03$$

$$\therefore V = 204.54 \text{ V}$$

b) Value of ' θ '

$$\hookrightarrow \tan \theta = \frac{PEL}{mu^2}$$

$$\text{or, } \theta = \tan^{-1} \left(\frac{1.76 \times 10^{11} \times 6818.18 \times 0.05}{(10^7)^2} \right)$$

$$\text{or, } \theta = \tan^{-1} \left(\frac{5.9 \times 10^{13}}{10^{14}} \right)$$

$$\text{or, } \theta = \tan^{-1} (0.599)$$

$$\therefore \theta \approx 30^\circ$$

c) resultant velocity at the point of emergence.

$\hookrightarrow$ Soln,

$$V = \sqrt{u^2 + \left(\frac{PE}{m} \times \frac{x}{u} \right)^2}$$

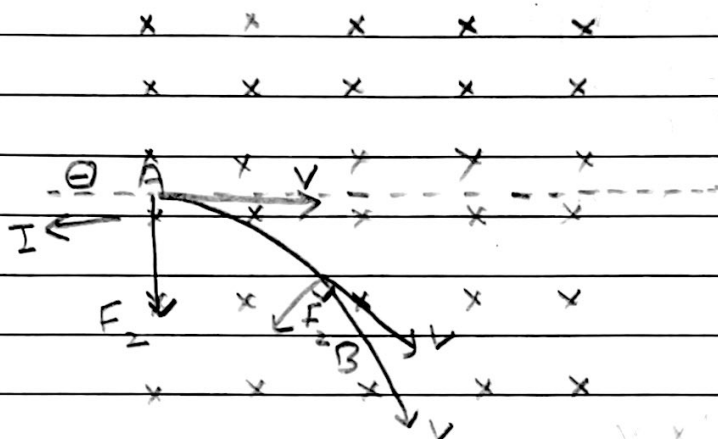
$$= \sqrt{(10^7)^2 + \left(\frac{1.76 \times 10^{11} \times 6818.18 \times 0.05}{10^7} \right)^2}$$

$$= \sqrt{10^{14} + 3.599 \times 10^{13}}$$

$$= \sqrt{1.35 \times 10^{14}}$$

$$= 1.16 \times 10^7 \text{ m/s}$$

- ⑤ An electron of mass 9.1×10^{-31} kg and charge 1.6×10^{-19} C is moving in a uniform magnetic field of 2×10^{-3} T perpendicularly as shown in figure.



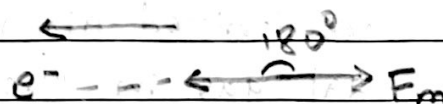
a) What is the path of electron and why?

→ The path of electron is circular path.

Since Magnetic Force acts continuously perpendicular to the velocity of particle, work done by force is zero and speed of particle remain constant. so, its direction changes. and the path of particle becomes circular.

b) What happens if the electron is passed by making 180° to this magnetic field?

→ If the electron is passed by making 180° to this magnetic field, electron goes straight in the direction parallel to magnetic field.



c) Find the speed of e^- and period of revolution if radius of path of electron is 20 cm.

↳ Soln

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$B = 2 \times 10^{-3} \text{ T}$$

$$\text{Given, } r = 20 \text{ cm} = 0.2 \text{ m}$$

Now,

$$r = \frac{mv}{qB}$$

$$0.2 = \frac{9.1 \times 10^{-31} \times v}{1.6 \times 10^{-19} \times 2 \times 10^{-3}}$$

$$\text{or, } \frac{6.4 \times 10^{-23}}{9.1 \times 10^{-31}} = v$$

$$\text{or, } v = 7 \times 10^7 \text{ m/s}$$

$$\therefore v = 7 \times 10^7 \text{ m/s}$$

⑦ What is meant by specific charge of an electron? It's value is constant for cathode rays but not for positive rays, why?

↳ The specific charge of an electron is the ratio of its charge to its mass (e/m).

For cathode rays, all particles are electrons (same charge and mass) regardless of the gas used in the discharge tube. So e/m is constant. For positive rays, the particles are ionized atoms/molecules of the residual gas in the tube, which differ in mass

and charge depending on the gas used; hence e/m varies for positive rays.

⑨ In what way Millikan's experiment lead to the quantum nature of electric charge?

→ Millikan found that the charge on different oil drops was always an integral multiple of a smallest value, $e = 1.6 \times 10^{-19} \text{ C}$, and no drop ever carried a fractional value of this charge.

This showed that electric charge is not continuous but exists in discrete packets (quanta), i.e. charge is quantized, given by $q = ne$, where n is an integer.